WEB DEVELOPMENT

Learning Groups: P107, P108, E101, P102

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Lesson 2 **learning topics**

- **2.0 HTML formatting**
- **2.1 HTML Styles**
- **2.2 HTML links**
- **2.3 HTML images**
- **2.4 HTML list**

Web editor for demonstration purposes use sublime text
But not limited

2.0 HTML Formatting Elements

HTML Formatting is a process of formatting text for better look and feel. HTML provides us ability to format text without using CSS. There are many formatting tags in HTML. These tags are used to make text bold, italicized, or underlined. There are almost 14 options available that how text appears in HTML and XHTML.

In HTML the formatting tags are divided into two categories:

- Physical tag: These tags are used to provide the visual appearance to the text.
- Logical tag: These tags are used to add some logical or semantic value to the text.

Element name	Description
<b>	This is a physical tag, which is used to bold the text written between it.
	This is a logical tag, which tells the browser that the text is important.
<i>	This is a physical tag which is used to make text italic.
	This is a logical tag which is used to display content in italic.
<mark>	This tag is used to highlight text.
<u>	This tag is used to underline text written between it.

<tt>	This tag is used to appear a text in teletype. (not supported in HTML5)
<strike>	This tag is used to draw a strikethrough on a section of text. (Not supported in HTM
<sup>	It displays the content slightly above the normal line.
<sub>	It displays the content slightly below the normal line.
<del>	This tag is used to display the deleted content.
<ins>	This tag displays the content which is added
<big>	This tag is used to increase the font size by one conventional unit.
<small>	This tag is used to decrease the font size by one unit from base font size.

Example illustrating Html text formatting elements.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Formatting Text in HTML</title>
</head>
<body>
    <p>This is <b>bold text</b>.</p>
    <p>This is <strong>strongly important text</strong>.</p>
    <p>This is <i>italic text</i>.</p>
    <p>This is <em>emphasized text</em>.</p>
    <p>This is <mark>highlighted text</mark>.</p>
    <p>This is <code>computer code</code>.</p>
    <p>This is <small>smaller text</small>.</p>
    <p>This is <sub>subscript</sub> and <sup>superscript</sup> text.</p>
    <p>This is <del>deleted text</del>.</p>
    <p>This is <ins>inserted text</ins>.</p><p>This is <b>bold text</b>.</p>
</body>
</html>
```

This is **bold text**.

This is **strongly important text**.

This is *italic text*.

This is *emphasized text*.

This is highlighted text.

This is `computer code`.

This is <small>smaller text.</small>

This is $_{\text{subscript}}$ and ^{superscript} text.

This is ~~deleted text~~.

This is inserted text.

This is **bold text**.

Difference between and <b> tag

Both and <b> tags render the enclosed text in a bold typeface by default, but the tag indicates that its contents have strong importance, whereas the <b> tag is simply used to draw the reader's attention without conveying any special importance.

Difference between and <i> tag

Similarly, both and <i> tags render the enclosed text in italic type by default, but the tag indicates that its contents have stressed emphasis compared to surrounding text, whereas the <i> tag is used for marking up text that is set off from the normal text for readability reasons, such as a technical term, an idiomatic phrase from another language, a thought, etc.

2.1 HTML Styles

Styles in **HTML** are basically rules that describe how a document will be presented in a browser. **Style** information can be either attached as a separate document or embedded in the **HTML** document.

There are 3 ways of implementing style in HTML :

1. **Inline Style**: In this method, the style attribute is used inside the HTML start tag.
2. **Embedded Style**: In this method, the style element is used inside the <head> element of the document.
3. **External Style Sheet**: In this method the <link> element is used to point to an external CSS file.

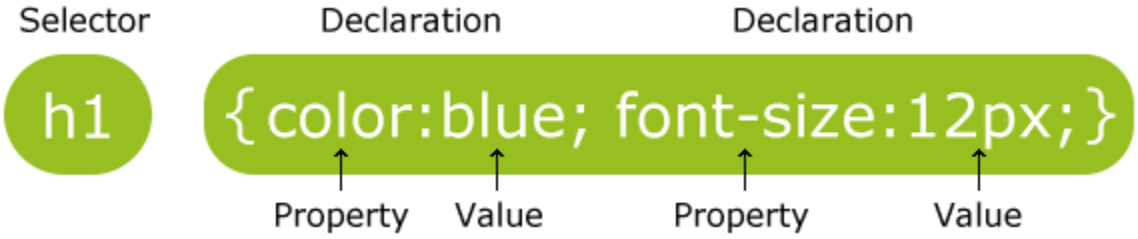
Note: The inline styles have the highest priority, and the external style sheets have the lowest. It means if you specify styles for your paragraphs in both *embedded* and *external* style sheets, the conflicting style rules in the embedded style sheet would override the external style sheet.

1. Inline Style

Inline Style: In Inline styling, the CSS rules are directly written inside the starting tag using the style attribute. The style attribute includes a series of CSS property and value pairs. Each '**property: value** 'pair is separated by a semicolon (;).

CSS Syntax

A CSS rule-set consists of a selector and a declaration block:



Inline style sheet css save as inlinestyle.html

```
<!DOCTYPE html>
<html>
<head>

</head>
<body>

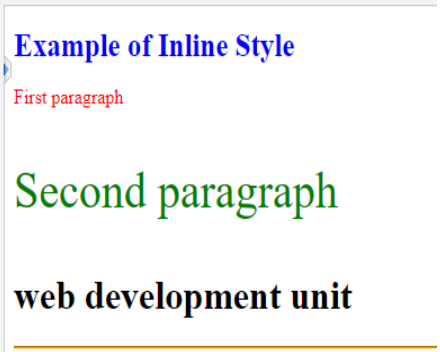
    <h1 style="color:Blue;font-size:25px;">
        Example of Inline Style
    </h1>

    <p style="color:red;">First paragraph</p>

    <p style="color:green;font-size:40px;">
        Second paragraph
    </p>
    <h1>web development unit</h1>
    <hr style="border-color:orange;">

</body>

</html>
```

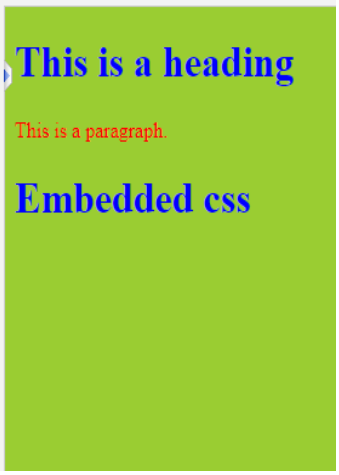


NB: ***it will change only the first h1 selector, changing the specific style where its applied. That's why in our output result first h1 colors blue and the other h1 remains by default black.***

2. Embedded Style /internal style save as internalstyle.html

Embedded style sheets are defined in the `<head>` section of an HTML document using the `<style>` tag. You can define any number of `<style>` elements inside the `<head>` section.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Using Embedded Style Sheet in HTML</title>
    <style type="text/css">
        body { background-color: YellowGreen; }
        h1 { color: blue; }
        p { color: red; }
    </style>
</head>
<body>
    <h1>This is a heading</h1>
    <p>This is a paragraph.</p>
  <h1>Embedded  css</h1>
</body>
</html>
```



NB: ***it will change all h1 elements to one color***

3.External Style Sheet:

External style sheets are the most flexible because with an external style sheet, you can change the look of an entire website by updating just one file.

You can attach external style sheets in two ways — *linking* and *importing*:

Here are the advantages of linking a CSS/external file to HTML:

- **Time-effective** — you only need to create a single CSS file to style all HTML files.
- **Faster load time** — the site will cache the CSS file for your visitors' next visit.
- **Improve SEO** — storing CSS styles in another file makes the HTML file more concise and readable by search engines.

Why not external Style-sheet?

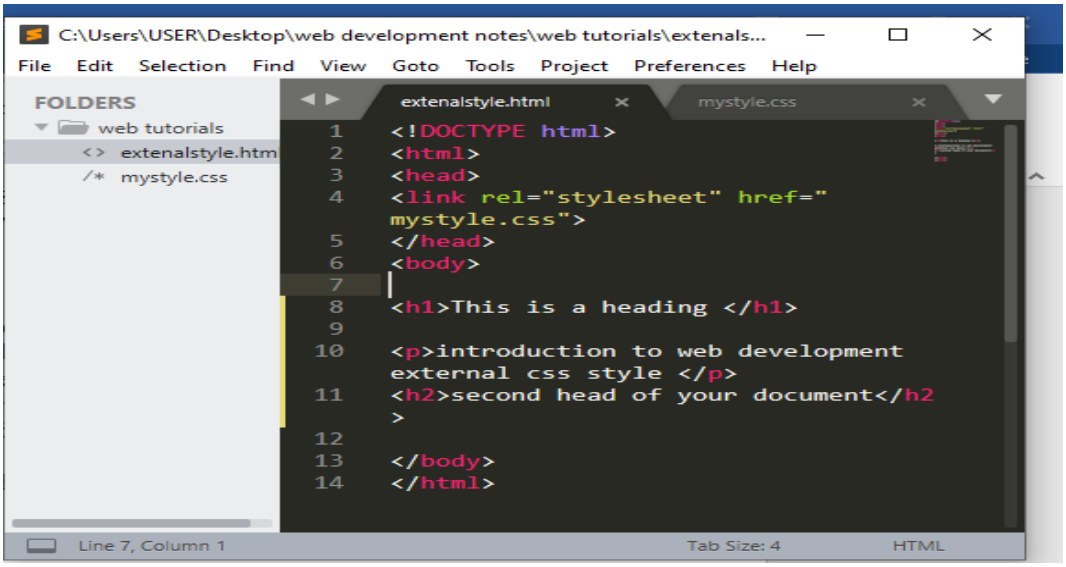
- If the style-sheet file has large data, then downloading the file takes time and it increases page download speed.
- For static site with limited pages and different design on each page, external style-sheets are harder to maintain.

NB: Example- one of the method using linking

```
<head>
    <link rel="stylesheet" href="css/style.css">
</head>
```

Link attribute is *embedded inside the head section*

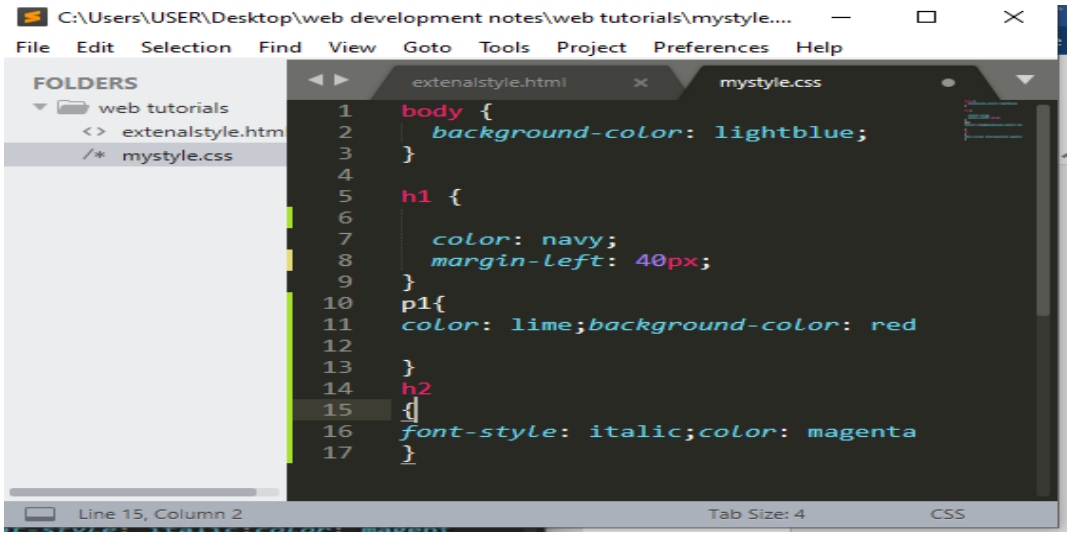
- **Link rel** — element defines the relationship between the file that hosts this command and the file defined in the **href** attribute. The standard value for this attribute is **stylesheet**.
- **type** — defines the content of the linked file. In this tutorial, set this attribute's value to **text/css**. However, you can skip it altogether if you're using HTML5
- **href** — specifies the location of the CSS file you want to link to the HTML. If the CSS file is in the same directory as the HTML file, you only need to enter the file name. Otherwise, you need to include the folder name in which you store the CSS file (example: **CSS/stylesheet.css**).



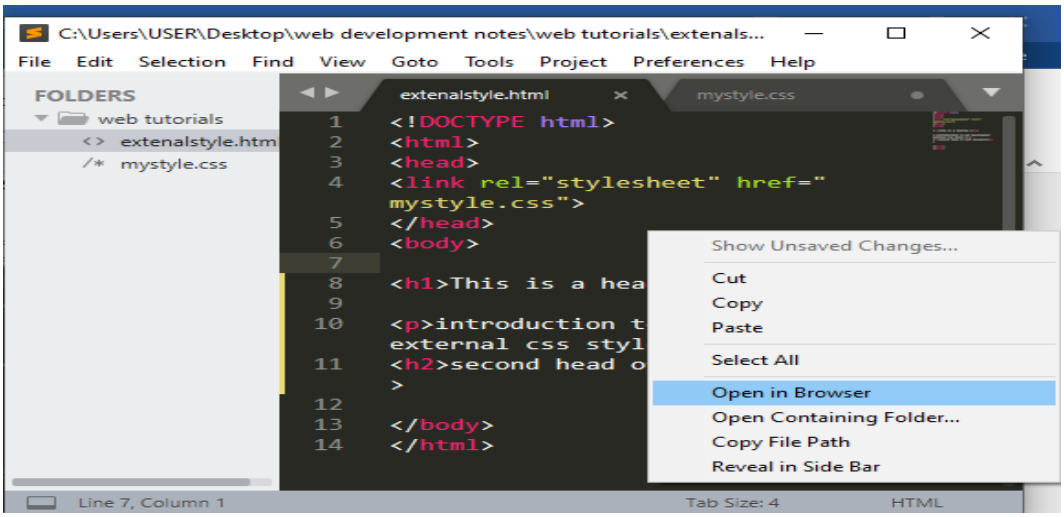
Step 1. Open sublime text > select file >open folder>select new folder>rename to your own name of choice.

Step 2. Inside named folder save document as externalstyle.html

Step 3. Open new tab **ctr+n**. write css rules and save as mystyle.css in the same folder



Step 4. Run the code using html file and right click >select open in browser



2.2 HTML LINKS

In HTML, links are defined with the **<a>** tag:

The **HTML anchor tag <a>** defines *a hyperlink that links one page to another page*. It can create hyperlink to other web page as well as files, location, or any URL.

Href (pronounced "h" – like the letter – "ref) is short for Hypertext Reference. This is a goofy name that really only means a URL (otherwise known as a web site address)

```
<a href = "....."> Link Text </a>
```

```
<a href="url">Link text</a>
```



Example

```
<a href="http://www.w3schools.com/html/">Visit our HTML tutorial</a>
```

Local Links

The example above used an **absolute URL** (A full web address).

A local link (link to the same web site) is specified with a **relative URL** (without http://www....)

HTML Links – Colors

When you move the mouse over a link, two things will normally happen:

- The mouse arrow will turn into a little hand
- The color of the link element will change

By default, a link will appear like this (in all browsers):

- An unvisited link is underlined and blue
- A visited link is underlined and purple
- An active link is underlined and red

HTML Links - The Target Attribute

The **target** attribute specifies where to open the linked document.

Example

```
<a href="http://www.w3schools.com/" target="_blank">Visit W3Schools! </a>
```

Types of Html links target attributes {blank, self, parent, top, frame name}

Target Value	Description
_blank	Opens the linked document in a new window or tab
_self	Opens the linked document in the same frame as it was clicked (this is default)
_parent	Opens the linked document in the parent frame
_top	Opens the linked document in the full body of the window

framename	Opens the linked document in a named frame

Example of target attribute

Edit This Code: See Result »

```
<!DOCTYPE html>
<html>
<body>

<a href="index.html" target="_blank">Visit our HTML tutorial!</a>

<p>If you set the target attribute to "_blank", the link will open in a new
browser window or tab.</p>

</body>

</html>
```

Result:

Visit our HTML tutorial!

If you set the target attribute to "_blank", the link will open in a new browser window or tab.

2.3 HTML IMAGES

images enhance visual appearance of the web pages by making them more interesting and colorful.

The <img> tag is used to insert images in the HTML documents. It is an **empty element** and contains attributes only. The syntax of the <img> tag can be given with:

```
<img src="url" alt="some_text">
```

Let's see an example of HTML image.

```
<!DOCTYPE>
<html>
<body>
<h2>HTML Image Example</h2>
<img src="good-morning.jpg" alt="Good Morning Friends"/>
</body>
</html>
```

HTML Image Example



Attributes of HTML image tag

The src and alt are important attributes of HTML img tag. All attributes of HTML image tag are given below

1) src

It is a necessary attribute that describes the source or path of the image. It instructs the browser where to look for the image on the server .

The location of image may be on the same directory or another server.

2) alt

The alt attribute defines an alternate text for the image, if it can't be displayed. The value of the alt attribute describes the image in words. The alt attribute is considered good for SEO (search engine optimization) prospective.

3) width

It is an optional attribute which is used to specify the width to display the image. It is not recommended now. You should apply CSS in place of width attribute .

4) height

It h3 the height of the image. The HTML height attribute also supports iframe, image and object elements. It is not recommended now. You should apply CSS in place of height attribute.

Use of height and width attribute with img tag

if we want to give some height and width to display image according to our requirement, then we can set it with height and width attributes of image.

```
<!DOCTYPE html>
<html>
<head>
        <title>Image tag</title>
</head>
<body>
<h2>HTML image example with height and width</h2>
<img src="https://static.javatpoint.com/htmlpages/images/animal.jpg"
height="180" width="300" alt="animal image">
</body>
</html>
```

HTML image example with height and width



NB: **How to get image from another directory/folder?**

```
<img src="E:/images/animal.png" height="180" width="300" alt="animal image">
```

E: - represents location from desktop drive use your preferred choice according to how you have saved your drive names. *You can omit width and height its optional*

Note: If src URL will be incorrect or misspell then it will not display your image on web page, so try to put correct URL

2.4 HTML – Lists

HTML offers web authors three ways for specifying lists of information. All lists must contain one or more list elements. Lists may contain –

- ** – An unordered list**. This will list items using plain bullets.
- ** – An ordered list**. This will use different schemes of numbers to list your items.
- **<dl> – A definition list**. This arranges your items in the same way as they are arranged in a dictionary.

HTML Unordered Lists

An unordered list is a collection of related items that have no special order or sequence. This list is created by using HTML **<ul>** tag. Each item in the list is marked with a bullet

```
<!DOCTYPE html>
<html>
<body>

<h2>Unordered List with Default Bullets</h2>

<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>

</body>

</html>
```

Unordered List with Default Bullets

- Coffee
- Tea
- Milk

Unordered HTML Lists - The Style Attribute

A **style** attribute can be added to an **unordered list**, to define the style of the marker:

Style	Description
list-style-type:disc	The list items will be marked with bullets (default)
list-style-type:circle	The list items will be marked with circles

list-style-type:square	The list items will be marked with squares
list-style-type:none	The list items will not be marked

```
<!DOCTYPE html>
<html>
<body>

<h2>Unordered List with Square Bullets</h2>

<ul style="list-style-type:square">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>

</body>

</html>
```

Unordered List with Square Bullets

- Coffee
- Tea
- Milk

Ordered HTML Lists - The Type Attribute

A **type** attribute can be added to an **ordered list**, to define the type of the marker:

Type	Description
type="1"	The list items will be numbered with numbers (default)
type="A"	The list items will be numbered with uppercase letters
type="a"	The list items will be numbered with lowercase letters
type="I"	The list items will be numbered with uppercase roman numbers
type="i"	The list items will be numbered with lowercase roman numbers

```
<!DOCTYPE html>
<html>
<body>

<h2>Ordered List with Numbers</h2>

<ol type="1">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>

</body>


</html>
```

Ordered List with Numbers

1. Coffee
2. Tea
3. Milk

Uppercase Letters:

```
<!DOCTYPE html>
<html>
<body>

<h2>Ordered List with Letters</h2>

<ol type="A">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>

</body>

</html>
```

Ordered List with Letters

A. Coffee
B. Tea
C. Milk

HTML Description Lists

HTML also supports description lists. A description list is a list of terms, with a description of each term.

The **<dl>** tag defines the description list, the **<dt>** tag defines the term (name), and the **<dd>** tag describes each term:

```
<!DOCTYPE html>
<html>
<body>

<h2>A Description List</h2>

<dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
</dl>

</body>

</html>
```

A Description List

Coffee
- black hot drink

Milk
- white cold drink